

A Full Banach-Caradus Answer for arXiv:2406.02330

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Source And Target

The source paper is L. Abadias, F. J. Gonzalez-Dona, and J. Oliva-Maza, *Universality arising from invertible weighted composition operators*, arXiv:2406.02330.

In Section 5 the authors ask the Banach-space version of their Hilbert-space universality theorem. Let

$$\mathcal{B} = H^p \quad \text{or} \quad \mathcal{B} = \mathcal{A}_\sigma^p, \quad 1 \leq p < \infty, \quad \sigma > -1,$$

and let uC_ψ be an invertible weighted composition operator whose symbol ψ is a hyperbolic automorphism of $\mathbb{D}$, with attractive and repulsive fixed points $a, b \in \mathbb{T}$. With the source notation

$$A^+ = \limsup_{z \rightarrow a} |u(z)|, \quad A^- = \liminf_{z \rightarrow a} |u(z)|, \quad B^+ = \limsup_{z \rightarrow b} |u(z)|, \quad B^- = \liminf_{z \rightarrow b} |u(z)|,$$

and with $\gamma = 1/p$ for H^p and $\gamma = (\sigma + 2)/p$ for $\mathcal{A}_\sigma^p$, assume

$$\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$$

and choose

$$\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma}.$$

The printed question asks whether $\ker(uC_\psi - \lambda I)$ is complemented and contains a subspace isomorphic to $\mathcal{B}$.

the following result:

Theorem 5.2. *Let uC_ψ be an invertible weighted composition operator with hyperbolic symbol ψ acting on $\mathcal{B}$ such that $\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$. Then, for every $\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma}$, $\ker(uC_\psi - \lambda I)$ is infinite dimensional and $uC_\psi - \lambda I$ is surjective.*

Observe that this result is not sufficient to obtain the universality for the operators $uC_\psi - \lambda I$ acting on the Banach spaces $\mathcal{B}$. Indeed, by Theorem 5.1 it would be enough to answer the following question, that remains open, at least for the knowledge of the authors:

Question. Let uC_ψ be an invertible weighted composition operator with hyperbolic symbol acting on $\mathcal{B}$, where $\mathcal{B}$ denotes the Hardy space H^p or the weighted Bergman space $\mathcal{A}_\sigma^p$, where $1 \leq p < \infty$ and $\sigma > -1$. Assume that $\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$ and let $\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma}$. Is $\ker(uC_\psi - \lambda I)$ a complemented subspace that contains a subspace which is isomorphic to $\mathcal{B}$?

We also observe that the analogous results of Proposition 3.2, Proposition 3.3 and Theorem 4.3 adapted to H^p and $\mathcal{A}_\sigma^p$ considerably improve previous results in the literature. Indeed, the spectrum of uC_ψ was given in [16, Theorem 4.9] provided that the function u belongs to the disc algebra (i.e. u has continuous extension to $\overline{\mathbb{D}}$, and then we write $A := A^+ = A^-$ and $B := B^+ = B^-$) and the

Result

Theorem 1. *Under the hypotheses above, $\ker(uC_\psi - \lambda I)$ is a complemented subspace of $\mathcal{B}$ and contains a closed subspace isomorphic to $\mathcal{B}$. Consequently $uC_\psi - \lambda I$ satisfies the Banach-space Caradus criterion.*

Proof Intuition

The source already proves that the eigenspace is infinite-dimensional and that $\lambda I - uC_\psi$ is onto. The first missing Banach-Caradus point is complementability; this is obtained by making the source's range-space decomposition linear and bounded, which gives a bounded right inverse.

The second missing point is largeness of the eigenspace. A hyperbolic automorphism acts freely and properly on $\mathbb{D}$, and the quotient $\mathbb{D}/\langle\psi\rangle$ is an annulus. If f is one nonzero eigenvector, then $f(h \circ \pi)$ is again an eigenvector for every analytic function h on the quotient annulus. In the Hardy case, the inherited norm on the annulus is a weighted Hardy norm with logarithmically integrable positive boundary weights, hence is isomorphic to the usual Hardy space on the annulus, and that space is isomorphic to $H^p(\mathbb{D})$. In the Bergman case, the already proved complementability and the standard identification $\mathcal{A}_\sigma^p \cong \ell^p$ imply that the whole infinite-dimensional kernel is itself isomorphic to ℓ^p , hence to $\mathcal{A}_\sigma^p$.

Standard Facts Used

Standard Fact 1 (Hyperbolic quotient). *Let ψ be a hyperbolic automorphism of $\mathbb{D}$. The cyclic group $\Gamma = \{\psi^n : n \in \mathbb{Z}\}$ acts freely and properly discontinuously on $\mathbb{D}$, and the quotient Riemann surface*

$$\Omega = \mathbb{D}/\Gamma$$

is conformally an annulus. If $\pi : \mathbb{D} \rightarrow \Omega$ is the quotient map, then the ψ -invariant holomorphic functions on $\mathbb{D}$ are exactly the functions $h \circ \pi$, $h \in \mathcal{O}(\Omega)$.

Standard Fact 2 (Weighted Hardy spaces on an annulus). *Let Ω be an annulus and let w_1, w_2 be positive weights on the two boundary components such that $w_j \in L^1$ and $\log w_j \in L^1$. Then*

$$H^p(\Omega, w) = \left\{ h \in \mathcal{O}(\Omega) : \sum_{j=1}^2 \int_{\partial_j \Omega} |h|^p w_j |d\zeta| < \infty \right\}$$

is isomorphic to the usual $H^p(\Omega)$, for $1 \leq p < \infty$. Moreover $H^p(\Omega) \cong H^p(\mathbb{D})$.

Reason for the fact. The annular outer-function theorem gives a zero-free holomorphic function F on Ω , with at most a harmless real power of the annular coordinate, such that $|F|^p$ has boundary values comparable to (w_1, w_2) . Multiplication by F identifies $H^p(\Omega, w)$ with the unweighted annular Hardy space. Finally, Laurent splitting identifies $H^p(\Omega)$ with two copies of $H^p(\mathbb{D})$, and $H^p(\mathbb{D}) \cong H^p(\mathbb{D}) \oplus H^p(\mathbb{D})$ by the even/odd decomposition. $\square$

Standard Fact 3 (Bergman structure). *For $1 \leq p < \infty$ and $\sigma > -1$, the weighted Bergman space $\mathcal{A}_\sigma^p$ is isomorphic, as a Banach space, to ℓ^p . The space ℓ^p is prime: every infinite-dimensional complemented subspace of ℓ^p is isomorphic to ℓ^p .*

Proof

Set

$$S_\lambda = \lambda I - uC_\psi.$$

The source paper proves in its Banach-space Section 5 that S_λ is surjective and that $\text{Ker } S_\lambda$ is infinite-dimensional.

Complementability

We first recall the complemented-kernel argument, since it supplies one of the two Banach-Caradus hypotheses.

For $\mu, \nu \geq 0$, set

$$\omega_{\mu,\nu}(z) = (a-z)^\mu(b-z)^\nu, \quad \mathcal{B}_{\mu,\nu} = \omega_{\mu,\nu}\mathcal{B}$$

with norm $\|g\|_{\mathcal{B}_{\mu,\nu}} = \|g/\omega_{\mu,\nu}\|_{\mathcal{B}}$. Choose $\mu, \nu > 0$ exactly as in the source proof of surjectivity:

$$|\lambda| > A^+(\psi'(a))^{\mu-\gamma}, \quad |\lambda| < B^-(\psi'(b))^{\nu-\gamma}.$$

The source argument shows that S_λ is boundedly invertible on $\mathcal{B}_{\mu,0}$ and on $\mathcal{B}_{0,\nu}$. Denote the two inverses by

$$R_\mu : \mathcal{B}_{\mu,0} \rightarrow \mathcal{B}_{\mu,0}, \quad R_\nu : \mathcal{B}_{0,\nu} \rightarrow \mathcal{B}_{0,\nu}.$$

Pick integers m, n with $m \geq \mu$ and $n \geq \nu$. The binomial identity used by the source can be read as a bounded linear splitting. Define

$$Pf = \frac{1}{(a-b)^{m+n}} \sum_{j=m}^{m+n} \binom{m+n}{j} (-1)^{m+n-j} (a-z)^j (b-z)^{m+n-j} f$$

and

$$Qf = \frac{1}{(a-b)^{m+n}} \sum_{j=0}^{m-1} \binom{m+n}{j} (-1)^{m+n-j} (a-z)^j (b-z)^{m+n-j} f.$$

Then $P, Q : \mathcal{B} \rightarrow \mathcal{B}$ are bounded, $P + Q = I_{\mathcal{B}}$, $\text{ran } P \subseteq \mathcal{B}_{\mu,0}$, and $\text{ran } Q \subseteq \mathcal{B}_{0,\nu}$. Hence

$$R = R_\mu P + R_\nu Q : \mathcal{B} \rightarrow \mathcal{B}$$

is bounded and $S_\lambda R = I_{\mathcal{B}}$. Therefore

$$\Pi = I_{\mathcal{B}} - RS_\lambda$$

is a bounded projection from $\mathcal{B}$ onto $\text{Ker } S_\lambda$. Thus the kernel is complemented.

The Hardy Case

Assume now $\mathcal{B} = H^p$, $1 \leq p < \infty$. By the source theorem, choose a nonzero $f \in \text{Ker } S_\lambda$. Let $\Gamma = \{\psi^n : n \in \mathbb{Z}\}$, let $\Omega = \mathbb{D}/\Gamma$, and let $\pi : \mathbb{D} \rightarrow \Omega$ be the quotient map.

For $h \in \mathcal{O}(\Omega)$, the lift $h \circ \pi$ is ψ -invariant, so

$$uC_\psi(f(h \circ \pi)) = (uC_\psi f)(h \circ \pi \circ \psi) = \lambda f(h \circ \pi).$$

Thus $f(h \circ \pi) \in \text{Ker } S_\lambda$ whenever it belongs to H^p . Define

$$E_f = \{h \in \mathcal{O}(\Omega) : f(h \circ \pi) \in H^p(\mathbb{D})\}, \quad \|h\|_{E_f} = \|f(h \circ \pi)\|_{H^p}.$$

The map

$$J : E_f \rightarrow \text{Ker } S_\lambda, \quad Jh = f(h \circ \pi),$$

is an isometry onto its range. It remains to identify E_f .

Let I_1, I_2 be fundamental boundary arcs for the action of Γ on the two components of $\mathbb{T} \setminus \{a, b\}$. Decomposing boundary measure over the disjoint arcs $\psi^n(I_j)$ and changing variables back to I_j gives

$$\|h\|_{E_f}^p = \sum_{j=1}^2 \int_{I_j} |h(\pi(\zeta))|^p W_j(\zeta) |d\zeta|,$$

where

$$W_j(\zeta) = \sum_{n \in \mathbb{Z}} |f(\psi^n(\zeta))|^p |(\psi^n)'(\zeta)|$$

in the boundary-change-of-variables sense. The weights are finite a.e. and integrable because $f \in H^p$. They are positive a.e. because a nonzero Hardy function has nonzero boundary values a.e. Finally, $\log W_j \in L^1(I_j)$: indeed W_j dominates one summand, and $\log |f| \in L^1(\mathbb{T})$ for nonzero $f \in H^p$.

Pushing these weights forward by π identifies E_f with a weighted Hardy space $H^p(\Omega, w)$ whose two boundary weights are positive, integrable, and have integrable logarithms. By the weighted annulus Hardy fact,

$$E_f \cong H^p(\Omega) \cong H^p(\mathbb{D}).$$

Therefore $J(E_f)$ is a closed subspace of $\text{Ker } S_\lambda$ isomorphic to H^p .

The Weighted Bergman Case

Assume now $\mathcal{B} = \mathcal{A}_\sigma^p$, $1 \leq p < \infty$, $\sigma > -1$. The source theorem gives $\dim \text{Ker } S_\lambda = \infty$, and the first part of the proof shows that $\text{Ker } S_\lambda$ is complemented in $\mathcal{A}_\sigma^p$.

By the standard atomic decomposition of weighted Bergman spaces, $\mathcal{A}_\sigma^p \cong \ell^p$. Since ℓ^p is prime, every infinite-dimensional complemented subspace of ℓ^p is isomorphic to ℓ^p . Transporting the projection onto $\text{Ker } S_\lambda$ through an isomorphism $\mathcal{A}_\sigma^p \cong \ell^p$, we conclude

$$\text{Ker } S_\lambda \cong \ell^p \cong \mathcal{A}_\sigma^p.$$

In particular the kernel contains a subspace isomorphic to $\mathcal{A}_\sigma^p$, namely the whole kernel.

Conclusion

For both $\mathcal{B} = H^p$ and $\mathcal{B} = \mathcal{A}_\sigma^p$, the kernel of $uC_\psi - \lambda I$ is complemented and contains a closed subspace isomorphic to $\mathcal{B}$. Since the source already proves that $uC_\psi - \lambda I$ is surjective, the Banach-space Caradus criterion applies. $\square$

Novelty And Literature Check

The run indexes contained only the earlier partial packet for arXiv:2406.02330. A fresh web search on 2026-07-03 for the arXiv id, the title, and phrases such as “kernel contains a subspace isomorphic”, “Banach weighted composition universal”, and “ uC_ψ Banach Caradus” found the source paper but no later paper explicitly answering the printed Section 5 question. The proof above is therefore recorded as a new full-solution candidate, subject to human review.

Human Review Recommendation

Review as a likely full solution. The complementability argument is a direct bounded-splitting refinement of the source proof. The main point to check is the Hardy annulus model: the boundary orbit decomposition should indeed give positive logarithmically integrable annular weights, so that the standard weighted Hardy factorization applies.

References

References

- [1] L. Abadias, F. J. Gonzalez-Dona, and J. Oliva-Maza, *Universality arising from invertible weighted composition operators*, arXiv:2406.02330.
- [2] S. R. Caradus, *Universal operators and invariant subspaces*, Proc. Amer. Math. Soc. 23 (1969), 526–527.
- [3] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.
- [4] K. Zhu, *Spaces of Holomorphic Functions in the Unit Ball*, Graduate Texts in Mathematics 226, Springer, 2005.
- [5] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Springer, 1977.